

DOM BASICS

1. What is the DOM and why does JavaScript need it?
 2. How is the DOM different from the original HTML file?
 3. Explain the DOM Tree using a real-life example.
 4. Is the DOM synchronous or asynchronous? Why?
 5. What happens to the DOM when JavaScript modifies an element?
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DOM SELECTION

6. What does `getElementById()` return if the ID does not exist?
7. Difference between `getElementsByName()` and `querySelectorAll()`.
8. Why does `getElementsByName()` return a live collection?
9. Predict the output:

```
const items = document.getElementsByTagName("p");
console.log(items.length);
```

10. Which selector is faster: `getElementById` or `querySelector`? Why?
11. Predict the output:

```
document.querySelectorAll(".box").forEach(e1 => console.log(e1));
```

12. Can you use array methods directly on `HTMLCollection`? Why?
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DOM MANIPULATION

13. How do you create an element and add text to it?
14. What happens if you append the same element twice?
15. Difference between `appendChild()` and `append()`.
16. Predict the result:

```
div.innerHTML = "<p>Hello</p>";
div.textContent = "World";
```

17. Why is innerHTML considered risky?
18. Difference between `textContent` and `innerText`.
19. What does `classList.toggle()` do?
20. Predict the output:

```
div.classList.add("a");  
div.classList.remove("a");  
console.log(div.className);
```

21. What is the difference between `setAttribute()` and direct property assignment?
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EVENTS

22. What is an event listener?
23. Why is `addEventListener` preferred over `onclick`?
24. What information does the event object contain?
25. Predict the output:

```
button.addEventListener("click", () => console.log("Button"));  
div.addEventListener("click", () => console.log("Div"));
```

26. Explain Event Bubbling using a real-world example.
27. How can you stop event bubbling?
28. What is Event Capturing?
29. What is the third argument in `addEventListener` used for?
30. What problem does Event Delegation solve?
31. Predict the output:

```
ul.addEventListener("click", e => console.log(e.target.tagName));
```

FORMS & INPUTS

32. What happens if `preventDefault()` is not used in form submission?
33. Difference between `submit` event and `click` on submit button.
34. What is the `input` event used for?

Dom & Browser Java Script (questions Only)

35. Predict the behavior:

```
input.addEventListener("input", () => console.log(input.value));
```

36. What is client-side validation?

37. Why is client-side validation not enough?

38. How do you validate required fields using JavaScript?

BROWSER APIs

39. What is Fetch API and why is it asynchronous?

40. Predict the output order:

```
fetch(url).then(() => console.log("Done"));  
console.log("Start");
```

41. Difference between localStorage and sessionStorage.

42. Where is localStorage data stored?

43. What happens to sessionStorage when the tab is closed?

44. Why are cookies sent with every HTTP request?

45. Difference between cookies and localStorage.

46. What permissions are required for Geolocation API?

47. What happens if user denies geolocation permission?

48. What is History API used for?

49. Predict what history.back() does.

50. Why are Browser APIs important for modern web applications?

NEXT STEP

After solving, say: 👉 **“Give me answers for Level 5 DOM questions”**

I will then provide: - Line-by-line explanations - Diagrams - Interview tips - Common mistakes